

Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

1 Introduction

Layered van der Waals solids are attractive thin-film materials because weak interlayer bonding enables exfoliation, heterostructure assembly, and growth on diverse substrates, while their electronic and transport properties remain sensitive to orientation and stacking.[1–3] Their structure is commonly measured with X-ray reflectivity, diffraction, and area-detector WAXS.[4–8] Depending on the substrate and growth conditions, the layer normals may align with the film normal while the crystallites remain nearly random in in-plane azimuth.[3] This uniaxially textured state is commonly called a two-dimensional (2D) powder: individual crystallites remain three-dimensional, but one crystallographic axis is preferentially aligned.[9, 10] At a single incidence angle, an area detector records specular peaks and diffuse intensity at the same time. These features provide complementary information about the film structure.[6–8]

Figure 1 illustrates this alignment and places the 2D-powder state between the single-crystal and 3D-powder limits in real, reciprocal, and detector space.

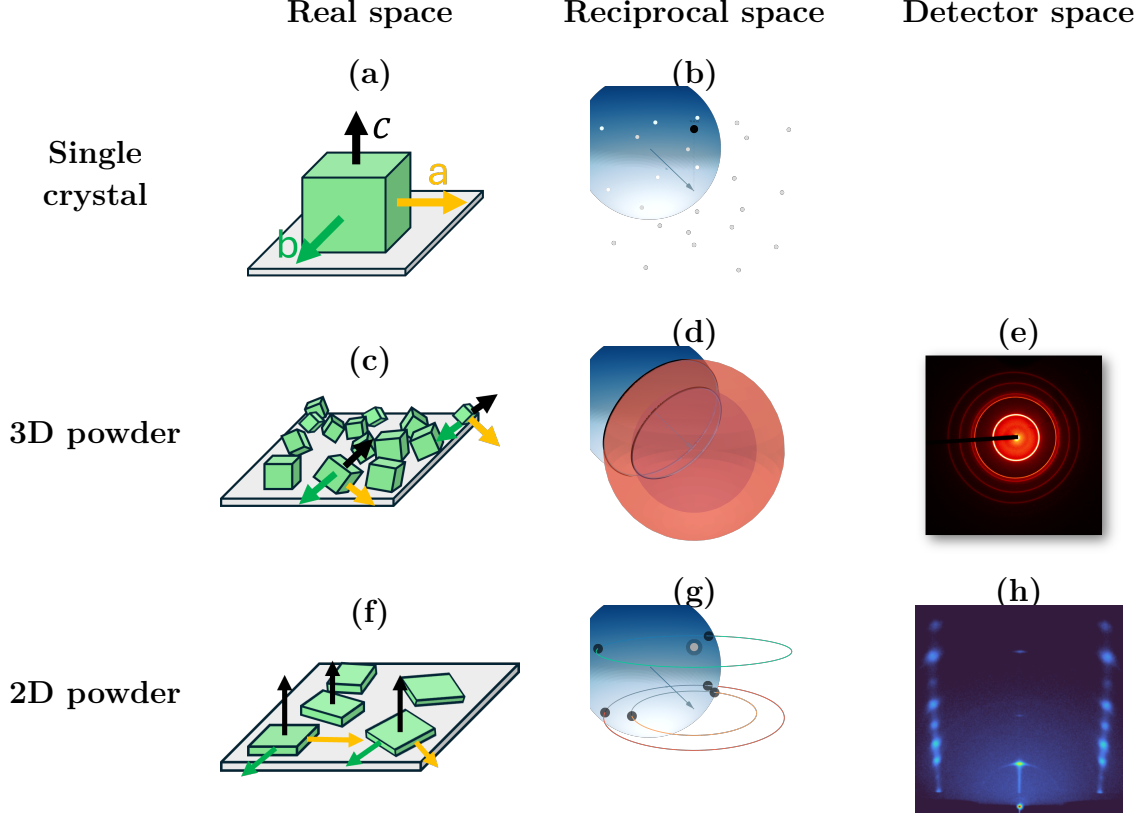


Figure 1: Diffraction across three orientational limits in real, reciprocal, and detector space. In the reciprocal-space panels (b,d,g), the blue Ewald sphere has radius $|\mathbf{k}_i| = 2\pi/\lambda$. Dark points or curves mark reciprocal-space intensity selected by that sphere. (a,b) A single crystal has isolated reciprocal-lattice points, so isolated intersections produce sparse detector spots. (c–e) Random 3D orientations spread a reflection over a spherical shell, whose intersection with the Ewald sphere maps to a Debye–Scherrer ring. (f–h) A 2D powder retains a common film normal but is random in in-plane azimuth. Off-specular reflections become rings about the film-normal reciprocal-space axis. Their Ewald-sphere intersections map to paired detector features when both fall within detector acceptance. The $00L$ intensity remains near the specular direction.

Figure 1 summarizes these orientational limits. Here $\mathbf{k}_i$ and $\mathbf{k}_f$ are the incident and exit wavevectors, and $\mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i$ is the momentum transfer. The Ewald sphere is centered at $-\mathbf{k}_i$, passes through the reciprocal-space origin, and has radius $|\mathbf{k}_i| = 2\pi/\lambda$. Elastic diffraction selects reciprocal-space intensity satisfying $|\mathbf{k}_i + \mathbf{Q}| = |\mathbf{k}_i|$, and each selected $\mathbf{Q}$ defines $\mathbf{k}_f = \mathbf{k}_i + \mathbf{Q}$.^[5] A single crystal supplies isolated reciprocal-lattice points, while random 3D orientations spread each reflection over a spherical shell. In a 2D powder, random in-plane azimuth spreads each off-specular reflection into a ring about the film-normal reciprocal-space axis, while $(00L)$ intensity remains near that axis.^[6, 9] The Ewald sphere generally intersects an off-specular ring at two points, which map to paired detector features.^[6, 7, 11] Mosaicity and other experimental effects broaden these ideal intersections into arcs and cap-like intensity while preserving the same elastic condition.

Figure 2 shows a representative measured PbI_2 pattern.

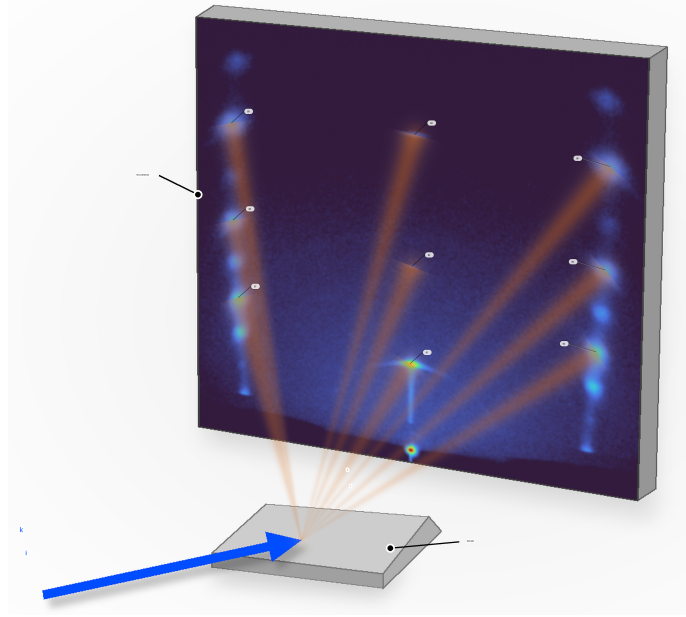


Figure 2: Illustration of a 2D-powder WAXS measurement using a measured PbI_2 sample. The incident beam strikes the film, and diffracted intensity is recorded directly on the area detector. Bright features indicate Bragg peaks, while the streaks between them and arcs illustrate how oriented crystallites, finite mosaic spread, and detector geometry distribute reciprocal-space intensity across the detector image.

Standard WAXS workflows often integrate area-detector images into one-dimensional profiles, an effective representation for nearly isotropic powders.[7, 8, 12, 13] For a 2D powder, retaining the radial and transverse detector directions keeps the specular and off-specular features as separate, complementary constraints. Stacking disorder is also commonly treated with one-dimensional interference models.[14–16] Full orientation-distribution-function refinements can represent arbitrary textures,[17–20] while more constrained cases can be represented by lower-dimensional preferred-orientation kernels[21, 22] or a small number of epitaxial variants.[23] The present work considers the 2D-powder limit. The approximately azimuthally symmetric measured features motivate axial symmetry about the film normal and a compact out-of-plane mosaicity distribution. We therefore use the off-specular arcs and specular-family profiles as complementary constraints on the same orientation model.

The optical boundary conditions follow established grazing-incidence formalisms,[7, 8, 24, 25] while the reciprocal-rod geometry and allowed-reflection locations build on reciprocal-space mapping and detector-space indexing for oriented films.[6, 11, 26, 27] For these data, wavelength bandwidth, beam divergence, refraction, sample misalignment, mosaicity, and stacking disorder can redistribute intensity in overlapping ways. We therefore evaluate them within one calculation. Source position and divergence are sampled from the measured Gaussian beam distributions, while wavelength is sampled from the retained spectrum. For each $\mathbf{k}_i$, the model constructs a structure function, applies the corresponding Ewald condition, and maps the accepted intensity to the calibrated detector. Applying identical integration regions to the measured and calculated detector intensities preserves the radial and transverse

information, so separate corrections to the collected data are not required.

Figures 8 and 9 compare the measured and calculated Bi_2Se_3 and Bi_2Te_3 profiles at a fitted incidence angle. The selected profiles agree in principal peak positions and apparent line shapes. These ordered-film test cases are followed by the staged refinement workflow and then the PbI_2 stacking-disorder extension.

2 Diffraction Model

We refer to the model as a forward model because it follows the physical sequence from ray to sample to detector. The incident beam on the sample is represented by weighted source states sampled in position, direction, and wavelength. Each incident ray is described by its incident angles (θ_i, ϕ_i) and wavevector magnitude $|\mathbf{k}_i| = 2\pi/\lambda$, so its state is written as $\mathbf{k}_i(|\mathbf{k}_i|, \theta_i, \phi_i)$. Sample composition and crystallinity determine a reciprocal-space structure function $S(\mathbf{k}_i, \mathbf{Q})$. Restricting this density to the Ewald sphere defined by $\mathbf{k}_i$ produces an Ewald surface decorated by the intensities in $\mathbf{Q}$. The corresponding $\mathbf{k}_f$ vectors carry the selected intensities toward the shared geometry of the chosen X-ray collection method, namely a 2D area detector.

2.1 Scattering and detector geometry

For one incident state, let $\mathbf{k}_i$ denote the incident wavevector and $\mathbf{k}_f$ the scattered wavevector. For wavelength λ , elastic scattering preserves their magnitudes,

$$|\mathbf{k}_i| = |\mathbf{k}_f| = \frac{2\pi}{\lambda}. \quad (1)$$

The momentum transfer is

$$\mathbf{Q} = \mathbf{k}_f - \mathbf{k}_i. \quad (2)$$

For an infinite, perfectly ordered crystal, an allowed reflection (h, k, ℓ) supplies

$$\mathbf{G}_{hkl} = h\mathbf{a}^* + k\mathbf{b}^* + \ell\mathbf{c}^*, \quad \mathbf{Q} = \mathbf{G}_{hkl}, \quad (3)$$

where $\mathbf{a}^*$, $\mathbf{b}^*$, and $\mathbf{c}^*$ include the factor 2π . The corresponding exit wavevector is

$$\mathbf{k}_f = \mathbf{k}_i + \mathbf{G}_{hkl}. \quad (4)$$

Equations (3) and (4) define the ideal integer Bragg-point limit. As developed later, replacing the integer order ℓ by a continuous coordinate L generalizes the same construction to a reciprocal rod, with the Bragg orders lying at $L = \ell$.

Figure 3 shows how an accepted exit direction reaches the area detector. The scattering angle 2θ and outgoing-ray azimuth ϕ_f locate $\mathbf{k}_f$, while detector distance D , axes x' and z' ,

normal $\hat{\mathbf{n}}_{\text{det}}$, and tilts β_D and γ_D determine the detector location. The detailed ray-plane and pixel-coordinate equations are given in the Supporting Information.

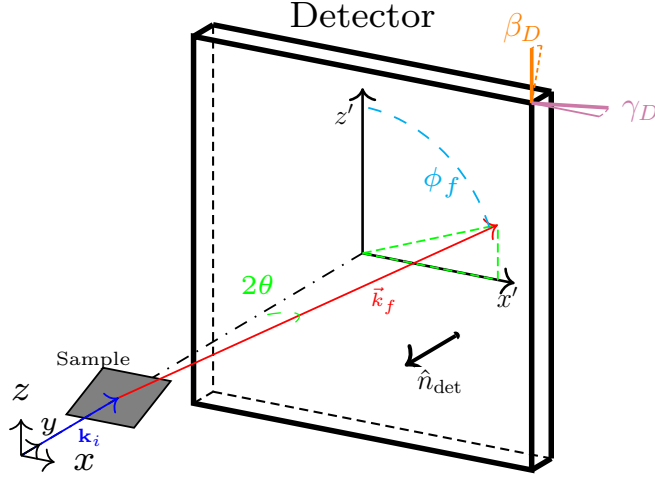


Figure 3: Laboratory and detector geometry. The incident wavevector $\mathbf{k}_i$ strikes the film, and a representative scattered wavevector $\mathbf{k}_f$ reaches the area detector at scattering angle 2θ and outgoing-ray azimuth ϕ_f . The detector axes x' and z' , normal $\hat{\mathbf{n}}_{\text{det}}$, distance D , and detector tilts define the ray-to-detector map.

Source position and propagation angles arising from divergence are sampled from the measured Gaussian beam distributions, while wavelength is sampled from the retained spectrum. The Supporting Information defines the weighted source-state construction. The external ray intersects the measured sample geometry, and refraction gives the internal vector $\mathbf{k}'_i$. Accepted exit rays must intersect the front face within the active detector bounds. Sample acceptance and detector acceptance are therefore consecutive geometric restrictions.

2.2 Structure Factor

For the hexagonal basal plane, every integer in-plane pair (h, k) belongs to the radial family

$$r \equiv h^2 + hk + k^2, \quad Q_R(r) = \frac{4\pi}{\sqrt{3}a} \sqrt{r}. \quad (5)$$

Members with the same r have the same in-plane radius. Their amplitudes are evaluated individually, and their intensity contributions are summed to represent scattering from the radial family. The family label is distinct from crystallographic multiplicity, and no second multiplicity factor is applied after this explicit sum.

For one (h, k) member, the crystallite-frame reciprocal rod is

$$\mathbf{G}_{hk}(L) = h\mathbf{a}^* + k\mathbf{b}^* + L\mathbf{c}^*. \quad (6)$$

The indices h and k are integers, while L is continuous. An integer Bragg order (h, k, ℓ) lies at $L = \ell$. The wavelength-dependent amplitude along the rod is

$$F_{hk}(L, \lambda_s) = \sum_a o_a f_a(|\mathbf{G}_{hk}(L)|, \lambda_s) T_a(\mathbf{G}_{hk}(L)) \exp[i\mathbf{G}_{hk}(L) \cdot \mathbf{r}_a]. \quad (7)$$

Here o_a , f_a , T_a , and $\mathbf{r}_a$ are the occupancy, atomic scattering factor, atomic displacement (Debye–Waller) factor, and Cartesian site position.

The complex amplitude F_{hk} is the structural input evaluated along the reference rod. The scalar object propagated through the remaining calculation is the continuous rod intensity $S_{hk}(L, \lambda_s)$. For an ordered finite film, S_{hk} is constructed from $|F_{hk}|^2$. $S_{hk}(L, \lambda_s) dL$ is the integrated structural intensity carried by a small segment of the unrotated (h, k) rod.

The cylinder construction follows directly from the definition of a 2D powder in Fig. 1(f–g): the crystallites share a mean film normal but are random in in-plane azimuth. If $\mathbf{C}$ maps the mean crystallite frame into the sample frame, rotating a point on the rod through azimuth β gives

$$\mathbf{Q}_{hk}^{\text{cyl}}(\beta, L) = \mathbf{C} \mathbf{R}_{\hat{\mathbf{c}}^*}(\beta) \mathbf{G}_{hk}(L), \quad 0 \leq \beta < 2\pi. \quad (8)$$

Thus, every off-specular rod sweeps out a cylinder about the film-normal reciprocal-space axis, while the specular rod remains unchanged. Members with the same r share the same geometric cylinder, although their amplitudes are evaluated separately. For off-specular cylinders with $Q_R > 0$, the circumference factor $1/[2\pi Q_R(r)]$ preserves the integrated intensity when a rod is distributed around that cylinder. This construction produces the paired off-specular trajectories in Fig. 1(h) while leaving the specular peaks behaving like single crystals.

2.3 Mosaicity

A two-dimensional powder is already random in in-plane azimuth. Mosaicity adds a distribution of crystallite tilts about the mean film normal. Its role is visible in the measured detector image in Fig. 4, where it explains features that are otherwise incompatible with a single incident vector and one perfectly aligned crystallite.

Figure 4 shows the $r = 0$ region of a Bi_2Se_3 image collected at the nominal incident angle $\theta_i = 5^\circ$. That incident condition places the Ewald intersection near the 003 condition. Surprisingly, intensity is also visible near the 006 location, at approximately twice the radial 2θ coordinate of 003. The 003 feature is also strongly anisotropic: its long extension follows the radial 2θ direction, whereas its transverse width follows the outgoing-ray azimuth ϕ .

For the 003 peak, the transverse ϕ width is primarily due to mosaicity, while the radial spread is better explained by wavelength dispersion. As discussed below, the unexpected 006 intensity motivates testing a broad component in the mosaic distribution. The bright feature near $L = 0$ is specular reflectivity, discussed separately in Sec. 2.5.

Radial and transverse structure of the Bi_2Se_3 $r = 0$ region

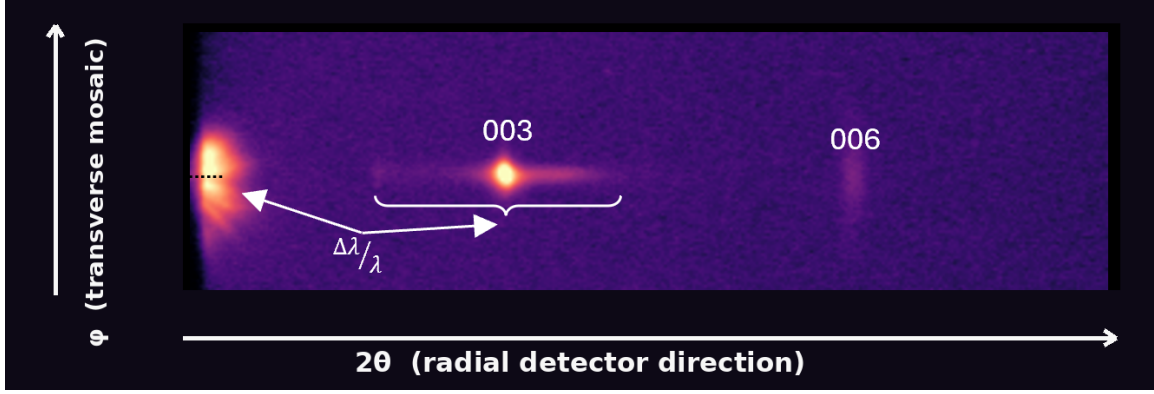


Figure 4: Measured $r = 0$ detector region from the 5° Bi_2Se_3 image. The nominal incident condition lies near 003, while intensity is also observed near the 006 location at approximately twice its radial 2θ coordinate. The 003 feature has a long radial extension along 2θ , whereas its transverse ϕ width is the cleaner constraint on crystallite tilt. The bright near-origin feature is near-critical specular reflectivity discussed in Sec. 2.5.

As shown below, off-specular rod-family arcs and the transverse ϕ width of off-condition $r = 0$ intensity probe different parts of the same orientation distribution. Strong off-specular arcs mainly constrain crystallites whose normals are close to the mean film normal. Weak $r = 0$ intensity farther from the specular direction is more sensitive to the small population at larger tilt.

To apply the crystallite mosaic tilt, we rotate each point on the powder cylinder through α about its local in-plane tilt axis,

$$\mathbf{Q}_{hk}(\alpha, \beta, L) = \mathbf{R}_{\hat{\mathbf{b}}_t(\beta)}(\alpha) \mathbf{Q}_{hk}^{\text{cyl}}(\beta, L), \quad (9)$$

where α is the crystallite-tilt magnitude and β is the cylinder azimuth. The local unit tilt axis is

$$\hat{\mathbf{b}}_t(\beta) = \mathbf{C} \mathbf{R}_{\hat{\mathbf{c}}^*}(\beta) \hat{\mathbf{b}}_t^0,$$

where $\hat{\mathbf{b}}_t^0$ is the reference in-plane unit axis. Thus, the tilt axis rotates with the cylinder azimuth β . At $\alpha = 0$, Eq. (9) reduces to the powder cylinder in Eq. (8).

The data were reproduced by a distribution with long tails,

$$\omega_w(\alpha; p_{\text{tail}}, \Gamma, \sigma) = p_{\text{tail}} L_w(\alpha; \Gamma) + (1 - p_{\text{tail}}) G_w(\alpha; \sigma), \quad 0 \leq p_{\text{tail}} \leq 1, \quad (10)$$

with the normalized orientation density

$$p_m(\alpha, \beta) = \frac{\omega_w(\alpha; p_{\text{tail}}, \Gamma, \sigma)}{2\pi}, \quad \int_0^\pi \int_0^{2\pi} p_m(\alpha, \beta) d\beta d\alpha = 1. \quad (11)$$

Here G_w is the Gaussian-like narrow core, L_w is the Lorentzian-like broad component, and p_{tail} is its weight. The two components share one center but have independent widths. The

Lorentzian-like form is not a claim that the physical distribution is uniquely Lorentzian.

Figure 5 explains the geometric sensitivity to a broad component. For $r \neq 0$, the strong detector arcs sample the orientation range near the nominal reciprocal vector. The transverse extent of off-condition $r = 0$ intensity probes larger tilts. Establishing that the broad component is necessary still requires a same-fit no-tail comparison and a fairly re-optimized Gaussian-only model, with residuals and uncertainty across the measured incident angles.

2.4 Ewald selection and mapping to the detector

The preceding sections introduce four distinct objects that together define the intensity before elastic selection. The structural rod intensity $S_{hk}(L, \lambda_s)$ is a scalar intensity density along the crystallite-frame rod containing the crystallographic amplitude. The location map $\mathbf{Q}_{hk}(\alpha, \beta, L)$ assigns each rod coordinate and crystallite orientation a point in three-dimensional reciprocal space. The orientation weight $p_m(\alpha, \beta)$ gives the probability density of that crystallite orientation and multiplies the rod intensity. The reciprocal-volume Jacobian $J_{\text{vol},hk}$ converts the weighted measure in (α, β, L) coordinates into a physical density per reciprocal-space volume. The selected intensity is then mapped to a 2D area detector.

For one source state s , a small pre-Ewald parameter element therefore carries

$$dI_{hk,s}^{\text{pre}} = S_{hk}(L, \lambda_s) p_m(\alpha, \beta) d\alpha d\beta dL \quad \text{at} \quad \mathbf{Q}_{hk}(\alpha, \beta, L). \quad (12)$$

The location map changes the associated volume element according to

$$d^3Q = J_{\text{vol},hk} d\alpha d\beta dL, \quad J_{\text{vol},hk} = \left| \det \frac{\partial \mathbf{Q}_{hk}}{\partial(\alpha, \beta, L)} \right|. \quad (13)$$

Consequently, the reciprocal-space intensity density is the sum of the weighted rod contributions divided by $J_{\text{vol},hk}$. In compact form,

$$\rho_{Q,s}(\mathbf{Q}) = \sum_{\substack{(h,k),j \\ \mathbf{Q}_{hk}(\alpha_j, \beta_j, L_j) = \mathbf{Q}}} \frac{S_{hk}(L_j, \lambda_s) p_m(\alpha_j, \beta_j)}{J_{\text{vol},hk,j}}. \quad (14)$$

S_{hk} supplies the structural intensity, $\mathbf{Q}_{hk}$ supplies its location, p_m supplies its statistical weight, and $J_{\text{vol},hk}$ preserves that intensity under the coordinate transformation.

After the incident optical boundary condition is applied, let $\mathbf{k}'_{i,s}$ be the internal incident wavevector for source state s , with $K_s = |\mathbf{k}'_{i,s}|$. Before elastic selection, the three coordinates (α, β, L) are independent: α and β set the crystallite orientation, while L sets the position along the rod. For a fixed source state, rod, and Ewald-intersection branch j , the Ewald condition supplies one scalar equation. Solving that equation removes one unknown by determining the compatible intrinsic rod coordinate $L = L_{hk,j,s}(\alpha, \beta)$:

$$\left| \mathbf{k}'_{i,s} + \mathbf{Q}_{hk}(\alpha, \beta, L) \right| = K_s, \quad \mathbf{Q}_{hk,j,s}^E(\alpha, \beta) = \mathbf{Q}_{hk}[\alpha, \beta, L_{hk,j,s}(\alpha, \beta)]. \quad (15)$$

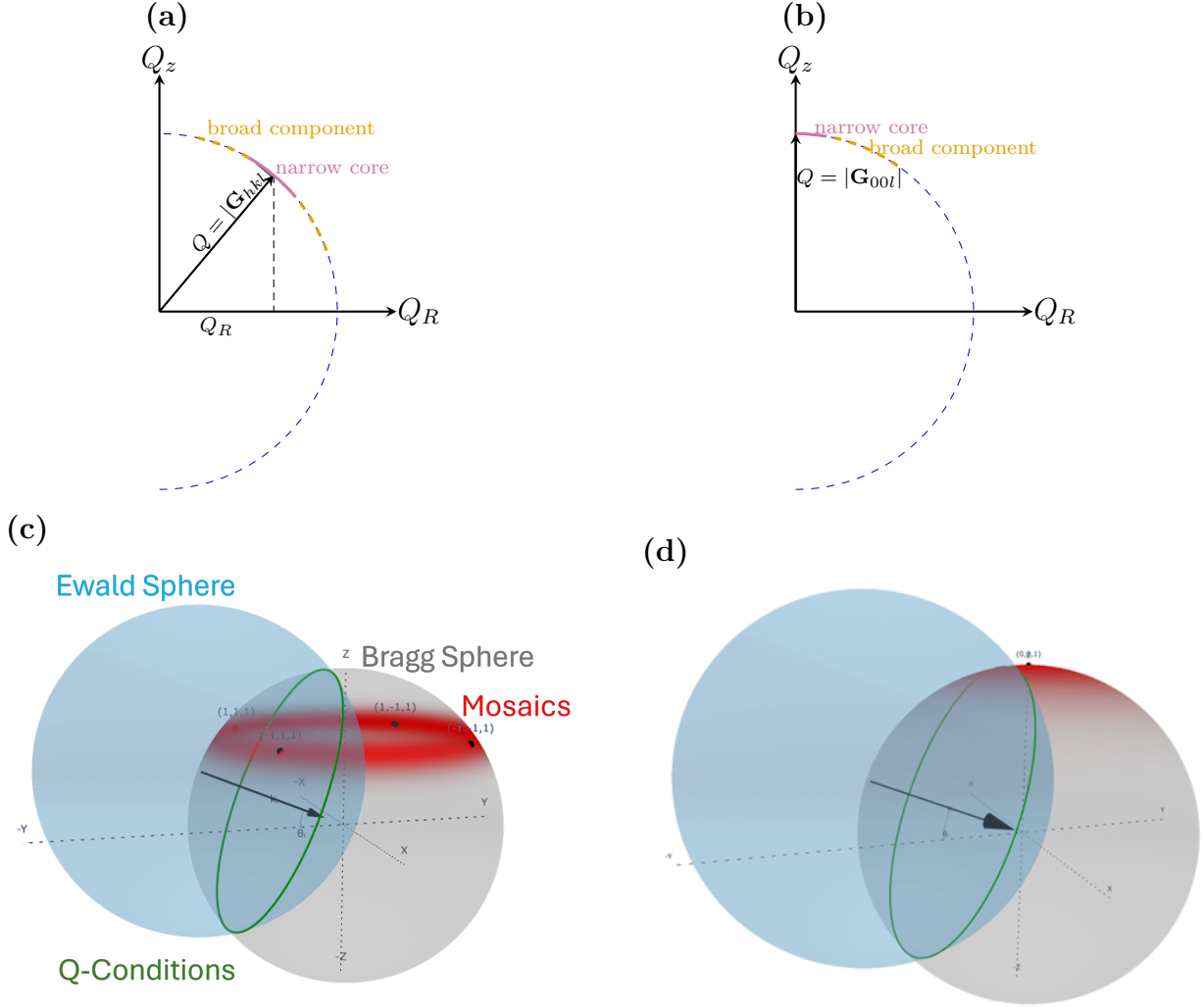


Figure 5: Geometric sensitivity to the narrow and broad crystallite-tilt components. (a) For $r \neq 0$, the reciprocal vector has radial and film-normal components Q_R and Q_z . The purple segment identifies the narrow core that predominantly controls the strong off-specular arc width, while the dashed orange extension identifies the broad, low-probability component. (b) For $r = 0$, the same orientation distribution begins on the film-normal axis. Intensity displaced in the transverse ϕ direction samples larger crystallite tilts and is therefore more sensitive to the broad component. This transverse mosaic effect is distinct from the radial 2θ streak through 003, which is assigned primarily to wavelength dispersion in Fig. 4. Opposite tilts fold onto the same $Q_R \geq 0$ locus. (c,d) Three-dimensional views place the corresponding orientation distributions relative to the Ewald-sphere intersection. The schematic illustrates sensitivity and does not replace a controlled mosaic-component comparison.

The Ewald restriction therefore reduces the three-coordinate reciprocal-space description to a two-coordinate surface chart: L becomes the dependent coordinate selected by elasticity, and (α, β) remain as the two independent parameters of a local Ewald-surface patch embedded in three-dimensional reciprocal space.

The intensity decorating that patch is obtained by evaluating the rod intensity at the selected L and weighting it by the crystallite-orientation density, the population of the individual (h, k) rod, and the source-dependent optical factors. Schematically,

$$\rho_{hk,j,s}^E(\alpha, \beta) \propto \frac{p_m(\alpha, \beta) S_{hk}[L_{hk,j,s}(\alpha, \beta), \lambda_s] W_{hk,j,s}}{J_{\text{vol},hk}}. \quad (16)$$

Here ρ^E is intensity per unit Ewald-surface area.

Each selected momentum transfer defines the internal exit vector $\mathbf{k}'_{f,s} = \mathbf{k}'_{i,s} + \mathbf{Q}^E$. The exit boundary condition converts this vector to the outgoing wavevector in air, and the calibrated detector geometry maps that ray to continuous detector coordinates. This final coordinate change introduces the detector Jacobian $J_{\text{det},s}$, which converts intensity per Ewald-surface area into intensity per detector area. So $J_{\text{vol},hk}$ first establishes the reciprocal-space density, and $J_{\text{det},s}$ then maps the selected surface density to the detector. Their net geometric contribution is $J_{\text{det},s}/J_{\text{vol},hk}$. The detector Jacobian contains the pixel solid angle, detector pose and distance, and the angular area change associated with exit refraction, so no additional solid-angle correction is applied.

Figure 6 visualizes this source-resolved sequence for the nominal Bi_2Se_3 incidence angle of 5° . The color on the internal-film Ewald sphere represents the physical intensity per unit Ewald-surface area. The planar and tilted-detector panels show the corresponding detector-coordinate density after the detector Jacobian is applied.

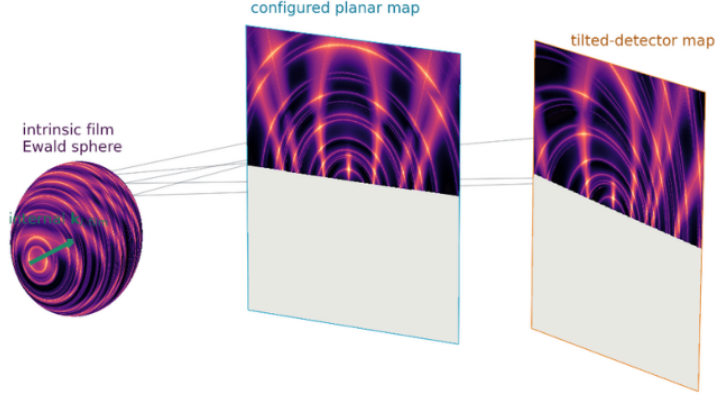


Figure 6: Source-resolved mapping from reciprocal-space intensity to detector coordinates for the nominal Bi_2Se_3 incidence angle of 5° . The left panel shows intensity per unit area on the internal-film Ewald surface after the rod, powder-cylinder, mosaic, and reciprocal-volume transformations. The center and right panels show the same selected intensity in the configured planar representation and on the tilted detector after exit refraction and the detector-coordinate Jacobian.

The calculation is repeated independently for every incident state, and the detector densities are summed with their source weights. The resulting image is in the same measurement space as the experimental detector image, so measured and calculated intensities are integrated over identical finite detector regions. Coordinates such as 2θ , ϕ , Q_R , and Q_z are downstream representations of this detector intensity. Here Q_z is the momentum-transfer component along the mean film normal and c is the crystallographic repeat length along that direction. The detector-derived projected coordinate used below is

$$L_{\text{proj}} = \frac{Q_z c}{2\pi}. \quad (17)$$

The crystallite-frame coordinate L identifies where $S_{hk}(L, \lambda_s)$ is evaluated. The projected coordinate L_{proj} is formed only after orientation, Ewald selection, and detector mapping. The two coincide only in the ideally aligned limit.

2.5 Reflectivity

The motivation for a separate specular treatment is visible directly in Fig. 4. In addition to the 003 and 006 diffraction features, the measured $r = 0$ profile contains a much brighter near-origin feature in the near-critical region. Its position and intensity scale identify a specular optical response rather than another finite-stack Bragg peak or independent evidence for a broad mosaic tail.

We therefore describe the measured $r = 0$ profile with two limiting calculations. In the

near-critical, low- Q_z region, a Parratt recursion accounts for refraction, absorption, evanescence, and multiple reflections at the interfaces.[4] Interfacial roughness modifies the Fresnel amplitudes within that Parratt branch through the Névot–Croce factor.[28] At higher Q_z , including the 003 Bragg region, the finite-stack kinematic rod intensity is used. The two branches are placed on the same detector-derived coordinate, background treatment, and intensity scale and are matched smoothly over a declared overlap interval. This practical stitching is the Parratt-to-kinematic crossover.

This construction is used here to keep the near-origin reflectivity distinct from the radial wavelength effect and transverse mosaicity. It is not an independent reflectivity refinement of density, optical constants, roughness, or thickness. Figure 7 illustrates the intended common-scale comparison. Direct measured and calculated arrays are required before it is used as quantitative evidence.

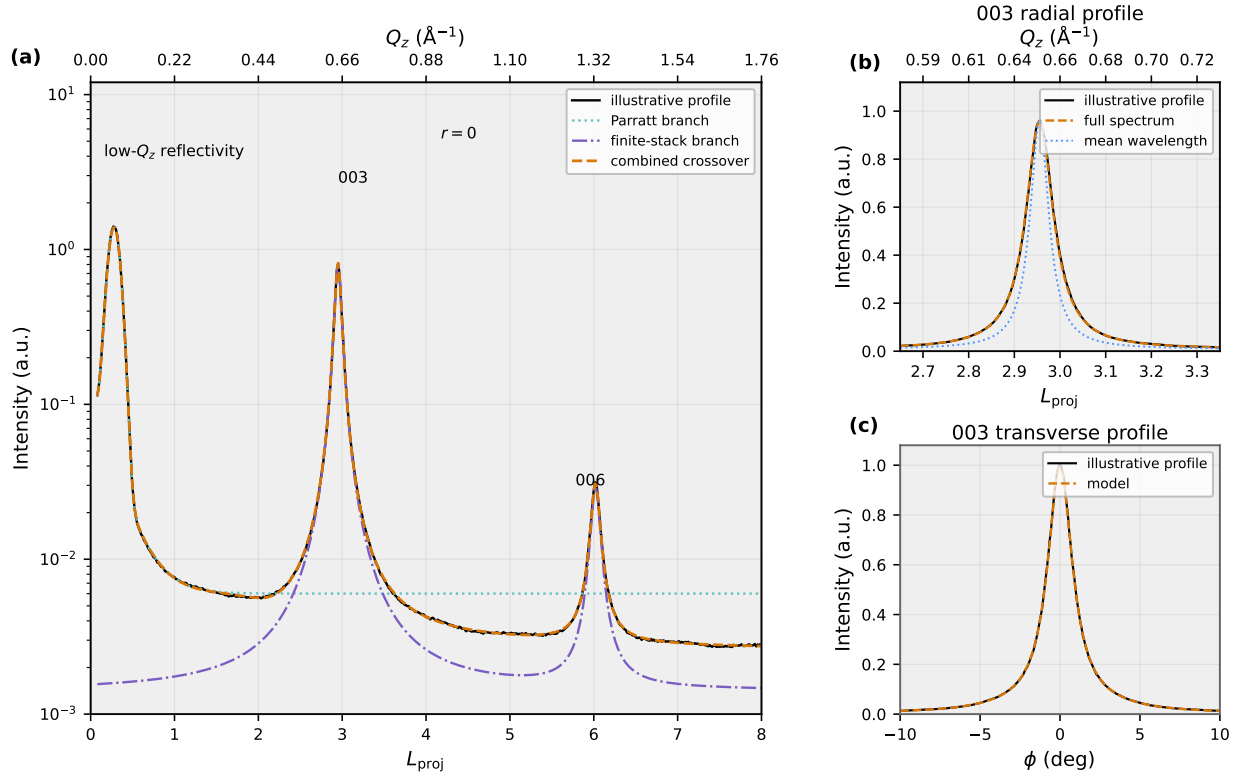


Figure 7: Illustrative construction of the stitched $r = 0$ specular response from the low- Q_z reflectivity region through the 003 Bragg peak. (a) The Parratt optical branch, finite-stack kinematic branch, and their smooth Parratt-to-kinematic crossover are shown on one coordinate transformation, background treatment, and intensity scale. Névot–Croce roughness is part of the Parratt branch rather than a separate contribution. (b) Expanded radial 003 profile illustrating the full wavelength-resolved calculation and a mean-wavelength control. (c) Transverse ϕ profile illustrating the mosaicity-sensitive direction. The lower horizontal axes use L_{proj} and the upper axes use Q_z . Direct measured and calculated arrays are required before this figure is used as quantitative evidence.

3 Validation: Bi_2Se_3 and Bi_2Te_3 films

Ordered Bi_2Se_3 and Bi_2Te_3 films provide validation. Detector images collected at incident angles of 5° , 10° , and 15° entered the staged refinement. Figures 8 and 9 show the 5° condition because it displays both the selected off-specular rod families and the $r = 0$ features clearly within one detector image while the others can be found in the supplemental materials.

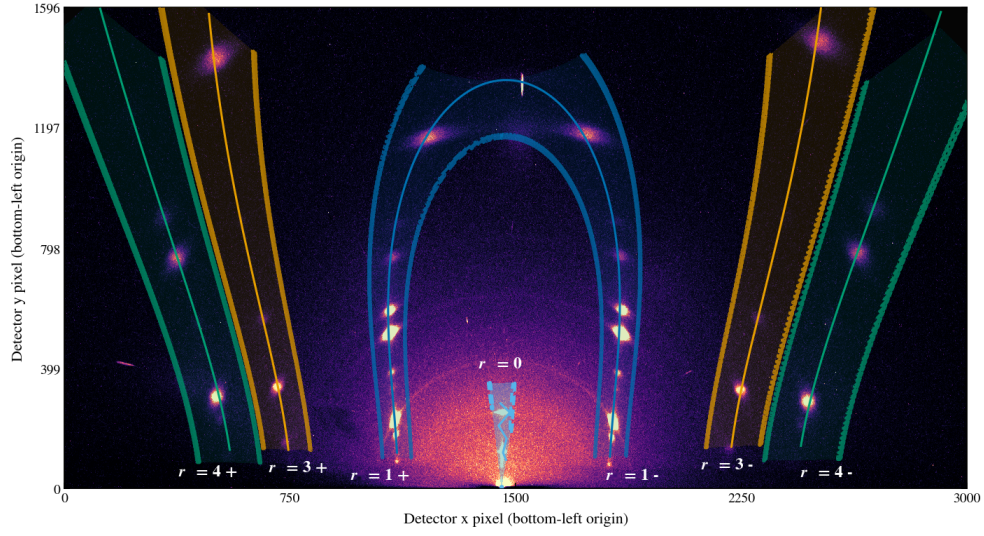
$r \neq 0$ rods are naturally described by Q space and integrated along Q_R as a function of L . Mosaic and beam sampling distort the projection to $\mathbf{Q}$ and thus the plotted curves therefore compare the same detector-derived observable rather than an ideal reciprocal-space line cut. Meanwhile, $r = 0$ rods are described best integrated along ϕ as a function of 2θ .

In the detector panels, solid colored curves mark calculated trajectory centerlines. Semi-transparent bands show the regions integrated for the profiles, and dashed outlines mark their boundaries. The central blue region follows the $r = 0$ family. Paired off-specular regions are the two detector-side intersections of one r family, identified by plus and minus signs. In the profile panels, black solid curves are measured data and orange dashed curves are the corresponding calculations. The horizontal coordinate is L , with integer markers denoting nominal Bragg conditions along the selected trajectory. Off-specular rows are plotted on a linear scale, while the $r = 0$ row is plotted logarithmically to display weak off-condition intensity. Along the $r = 0$ trajectory, the radial L or 2θ line shape contains the wavelength convolution, whereas the transverse ϕ width provides the cleaner mosaic constraint.

Mosaicity, beam divergence, wavelength spread, incidence-angle uncertainty, calibrated detector mapping, and finite integration width all broaden the effective scattering condition. The projected coordinate should therefore not be interpreted as an exact cut through the reciprocal space of one perfect crystallite. The comparison remains direct because the same coordinate transformation and finite integration operation are applied to the calculated and measured detector quantities.

The off-specular and $r = 0$ features play complementary roles. The off-specular arcs test the two-dimensional-powder reciprocal-space construction and calibrated detector geometry. Their transverse widths and asymmetries constrain the effective narrow mosaic component together with beam divergence and finite integration width. For the $r = 0$ family, the radial 2θ or L shape is sensitive to the wavelength distribution and the specular intensity envelope, while the transverse ϕ width is the cleaner probe of the broad orientation component. The full-spectrum versus mean-wavelength comparison in Sec. 2.4 isolates the radial 003 streak from the mosaic-sensitive transverse width. The displayed overlays show that the fitted model can represent the selected feature locations and apparent profile shapes. A controlled Gaussian-only comparison is still required before the broad component can be called necessary or uniquely Lorentzian.

Measured detector image and selected trajectory bands



Matched projected profiles

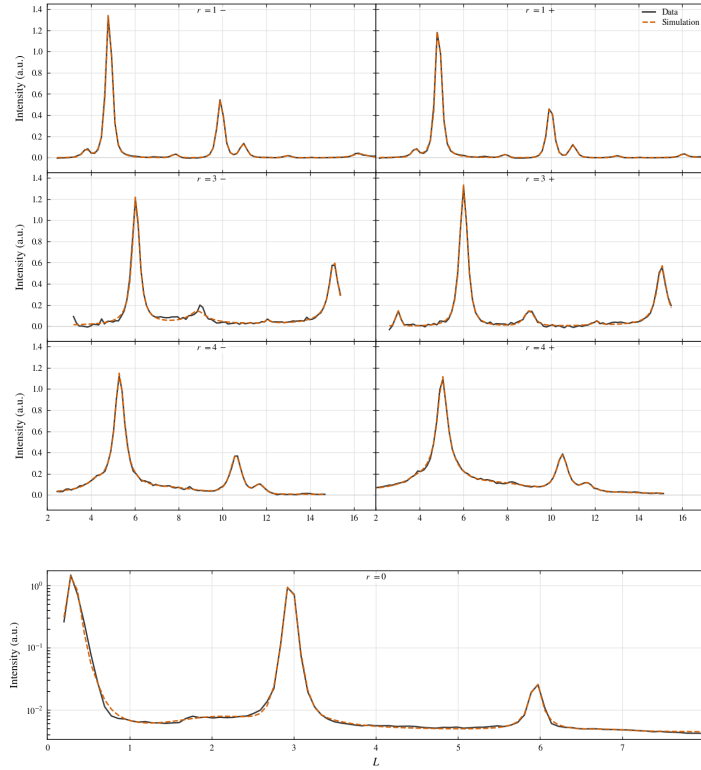
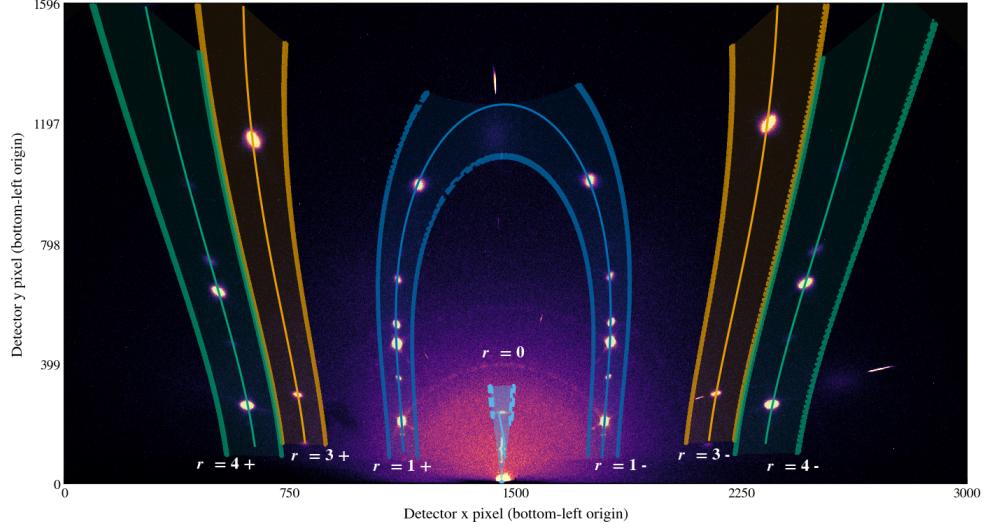


Figure 8: Bi_2Se_3 detector-space comparison at $\theta_i = 5^\circ$. The upper panel is the measured detector image with calculated trajectory centerlines, finite integration regions, and their boundaries. Labels r identify radial rod families, while plus and minus signs distinguish the two detector-side intersections of one off-specular family. The lower panel compares measured black curves with calculated orange dashed curves generated using the same coordinate map, integration limits, and area normalization. The figure tests trajectory placement and apparent profile line shape. In the $r = 0$ family, radial broadening is evaluated with the wavelength distribution, while the transverse ϕ width is used to constrain mosaicity. Separate family amplitudes are not a common-scale relative-intensity test, and the $r = 0$ row does not establish a unique broad-component functional form.

Measured detector image and selected trajectory bands



Matched projected profiles

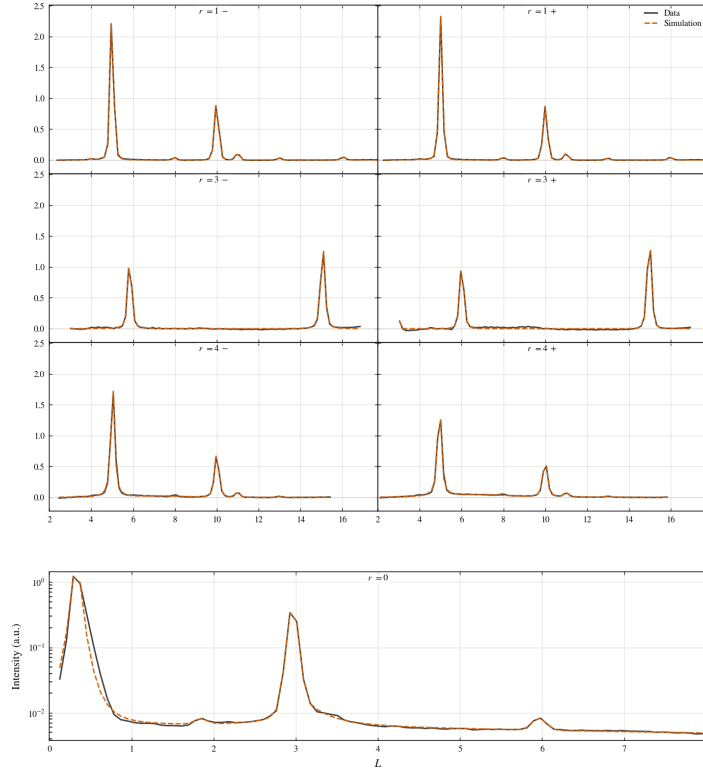


Figure 9: Bi_2Te_3 detector-space comparison at $\theta_i = 5^\circ$, using the conventions of Fig. 8. The upper panel shows the measured detector image with calculated trajectories and finite integration regions. The lower panel compares detector-derived profiles formed with the same coordinate transformation, finite integration, and area normalization. The principal peaks occur at the same projected locations and have similar apparent line shapes within the displayed fitted condition. Separate family amplitudes are not interpreted on one common physical scale, and the $r = 0$ profile is not proof of a unique broad mosaic component.

4 Refinement workflow

The refinement is staged rather than fully global from the beginning. Many parameters affect similar detector-space observables: detector distance shifts peak positions, sample alignment changes the apparent incident condition, mosaicity changes profile widths and off-condition intensity, and structure factors change relative peak heights. A simultaneous fit can therefore hide physical errors by compensating one parameter group with another. The staged workflow fixes the instrumental quantities first, then introduces sample orientation, mosaicity, ordered intensity, and finally stacking disorder where needed.

The first distinction is between instrumental parameters and sample parameters. Instrumental parameters describe the beam and detector and should be shared across samples collected under the same experimental configuration. Sample parameters describe the film: incidence condition, sample offset, mosaic distribution, structural intensity, and, for PbI_2 , stacking disorder. The refinement sequence in Table 1 uses the detector-space observable most sensitive to each group before that group is used in the final image projection.

Table 1: Staged refinement workflow. Steps 1–5 are used for the ordered Bi_2Se_3 and Bi_2Te_3 test cases. Step 6 is the added PbI_2 extension for stacking-fault diffuse scattering. All sample images are dark-subtracted before the listed observables are formed. The final measured and calculated profiles use the same calibrated coordinate map, finite integration limits, and normalization.

Step	Stage	Data	Parameters refined	Primary information
1	Beam characterization	Direct-beam images at several detector distances	Beam widths w_x, w_z and divergences $\Delta\theta_x, \Delta\theta_z$	Incident phase space
2	Detector geometry calibration	Powder calibrant image	Detector distance D , detector tilts (β, γ) , and beam center (x_0, y_0)	Detector mapping
3	Sample alignment and goniometer misalignment	Indexed reflection centers from sample images	Sample geometry (θ_i, z_B, z_S) , in-plane rotation χ , goniometer misalignment (α, ψ)	Detector-space peak positions
4	Mosaicity/profile refinement	Selected off-specular ROIs, transverse ϕ profiles, and off-condition $r = 0$ regions	Narrow-core plus broad-component mosaic parameters	Off-specular arc widths, transverse ϕ widths, and off-condition specular intensity
5	Ordered-film comparison	Selected detector-space Bragg regions and Q_z projections from fixed- r trajectories	Image-wise nuisance scales and constrained ordered-intensity parameters	Selected Bragg-feature locations and apparent line shapes
6	PbI_2 diffuse extension	Fixed- Q_R diffuse regions, ordered-baseline residuals, and projected Q_z profiles	Stacking-fault probabilities or layer-pair correlation parameters	Diffuse intensity along Q_z and separation of ordered and disorder-derived scattering

The instrumental stages fix where an ideal reflection should appear before any sample-specific physics is refined. Direct-beam images define the incident phase space, including beam width and angular divergence. A powder calibrant fixes the detector mapping: distance, beam center, and detector tilts. These two stages are intentionally separated from

sample fitting because an incorrect detector geometry can otherwise be absorbed into apparent sample tilt, mosaic broadening, or shifted reciprocal-space trajectories.

The alignment stage is the part of the workflow most similar to detector-space indexing. Indexed sample reflections are used to determine how the film sits in the beam, fixing the nominal incident angle, sample offsets, in-plane rotation, and goniometer misalignment well enough that calculated Bragg-rod trajectories can be overlaid on the detector. This establishes where allowed reflections should appear, analogous to detector-space indexing tools such as indexGIXS.[27] Only after this geometric step does it make sense to refine the mosaic distribution, because mosaicity should explain profile shape and off-condition intensity rather than compensate for a misplaced trajectory.

The mosaicity stage constrains the shared narrow-core plus broad-component angular distribution. Local off-specular profiles constrain the narrow core. Within the $r = 0$ family, transverse ϕ profiles constrain crystallite tilt, while the radial 2θ profile is held out of the mosaic interpretation until the wavelength distribution is fixed. With geometry and mosaicity held fixed, the full-spectrum and mean-wavelength calculations are then compared directly. The successful control assigns the radial 003 streak to wavelength dispersion rather than to the broad mosaic component. The ordered-film comparison applies the same calibrated coordinate map and finite profile integration to the measured and calculated detector quantities. The present main-text figures use this stage to compare feature locations and apparent profile shapes. Quantitative relative-intensity refinement requires a documented common normalization and retained direct arrays.

For PbI_2 , the ordered workflow is not discarded. It provides the geometry, projection, and mosaic baseline needed before diffuse scattering can be interpreted. The additional stacking-disorder stage asks whether a physically defined layer-correlation model accounts for the remaining diffuse intensity along fixed- Q_R cylinders. This keeps the PbI_2 result tied to the same detector-space framework while making clear which extra parameters belong specifically to stacking faults.

Implementation details, including weighted source-state quadrature, analytic Ewald restriction, detector-coordinate integration, parameter-correlation plots, and propagated uncertainty from the 2θ - ϕ and Q transformations, are documented in the Supporting Information. The main-text role of the workflow is to make the logic of the refinement reproducible. Instrumental geometry is determined first, sample orientation second, mosaic and profile effects third, and ordered or diffuse intensity differences are interpreted as structural information only after those stages are fixed.

5 Results: diffuse scattering and stacking disorder in PbI_2

Figure 10 shows a CVD-grown PbI_2 film and a crop spanning the $r = 0$ and $r = 1$ reciprocal rods. The $r = 1$ rod contains measurable intensity between the integer Bragg maxima,

whereas the $r = 0$ rod is dominated by sharp peaks and finite-thickness oscillations. This rod-dependent redistribution is the experimental motivation for the stacking-disorder model. In a two-dimensional powder, random in-plane orientation maps each (h, k) rod onto a reciprocal-space cylinder at fixed Q_R . An infinite ordered crystal gives intensity at discrete crystallite-frame L values, while a finite ordered stack already produces a continuous $S_{hk}(L)$ with finite widths and fringes. Stacking faults do not create these rods. They change the correlations among trilayers and redistribute intensity within the existing continuous rod profile.

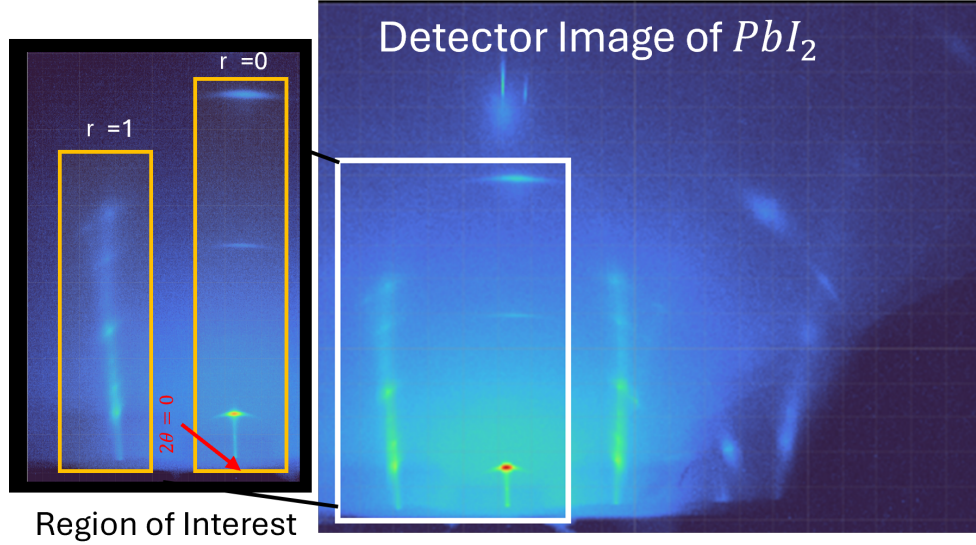


Figure 10: Measured PbI_2 diffraction image (right) and an enlarged region of interest (left). The direct-beam position defines the vertical $r = 0$ trajectory. Diffuse intensity is visible between the ordered maxima on the neighboring $r = 1$ rod, while the $r = 0$ profile is contrast-matched to the basal-plane shifts included in the stacking model.

The contrast between the two rods identifies the structural information carried by the diffuse intensity. A forward basal-plane shift $\mathbf{s} = (1/3, 2/3)$ multiplies the amplitude of one (h, k) rod by

$$\omega_{hk} = \exp[2\pi i(hs_x + ks_y)] = \exp\left[\frac{2\pi i}{3}(h + 2k)\right]. \quad (18)$$

For $h = k = 0$, $\omega_{00} = 1$, and the two trilayer orientations have the same $00L$ amplitude in the present model. The $r = 0$ rod is therefore insensitive to the included basal-plane shifts and orientation changes: faults can be present, but they are contrast-matched on that rod. Off-specular rods retain the lateral phase contrast, so the same faults change their pair interference and transfer intensity from ordered maxima into the diffuse regions between them.

5.1 From stacking faults to rod intensity

The structural unit is one I-Pb-I trilayer. Its state is described by a basal-plane position $\rho \in \{0, 1, 2\}$ and an orientation $\sigma \in \{+, -\}$. The ordered 2H, 4H, and 6H parents are deterministic sequences of these six possible layer states. Figure 11 renders the corresponding cycles as projected I-Pb-I stacks. Read from bottom to top, they are $0F_+ \rightarrow 0F_+$ for 2H, $0F_+ \rightarrow 1F_- \rightarrow 0F_+$ for 4H, and $0F_+ \rightarrow 1F_+ \rightarrow 2F_+ \rightarrow 0F_+$ for 6H. The existence and crystallographic stacking sequences of these parent polytypes are established experimentally.[29, 30]

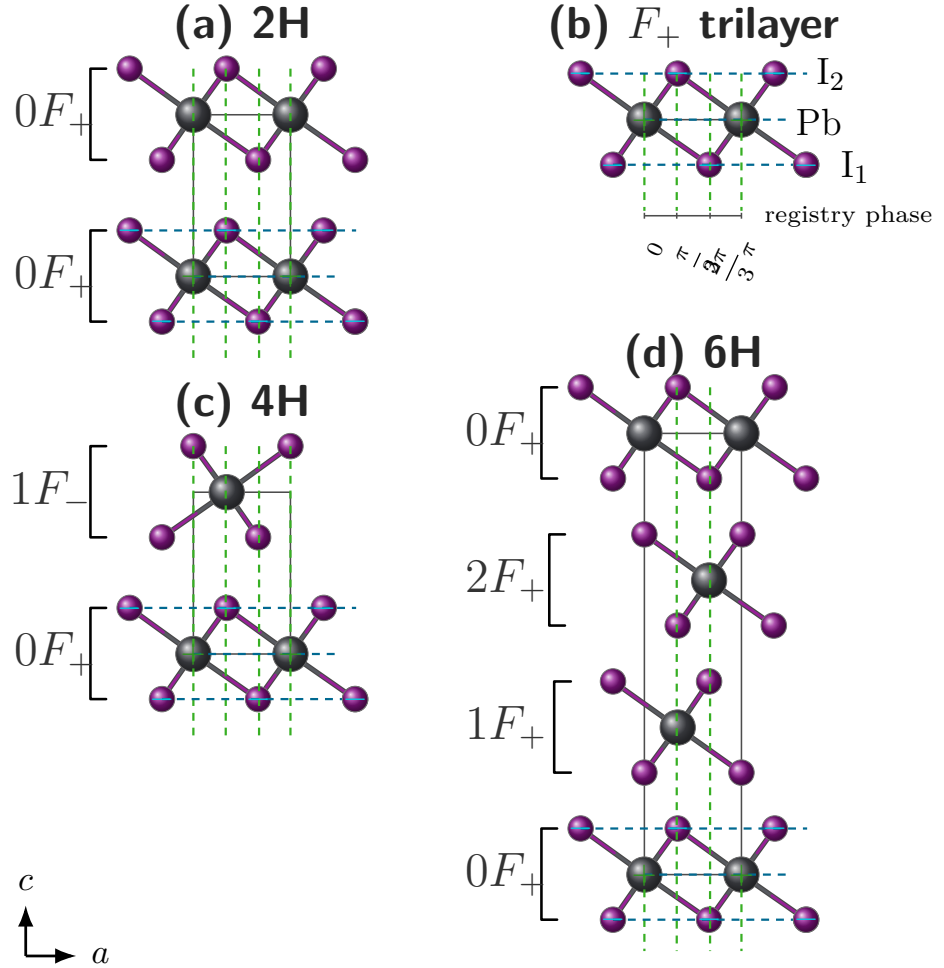


Figure 11: Deterministic PbI₂ parent cycles used in the transition model. (a) 2H repeats $0F_+$. (b) The isolated F_+ trilayer defines the local orientation. (c) 4H alternates $0F_+$ and $1F_-$. (d) 6H advances $0F_+ \rightarrow 1F_+ \rightarrow 2F_+ \rightarrow 0F_+$. In a state label ρF_σ , ρ is the basal-plane position and σ is the trilayer orientation.

Adjacent trilayers are connected by five allowed interface events. The event a leaves the basal-plane position and orientation unchanged. The events b_+ and b_- shift the next trilayer forward or backward without changing its orientation, while d_+ and d_- combine those shifts

with an orientation change. A change of orientation without a basal-plane shift is excluded. The parent events are a for 2H, d_+ for 4H, and b_+ for 6H. Reversing the shift direction gives the symmetry-related handed counterparts.

A stacking fault is represented as a local departure from the selected parent event. For a parent-rich population $j \in \{2H, 4H, 6H\}$, the equal-alternative template is

$$p_j(\text{parent event}) = 1 - \epsilon_j, \quad p_j(\text{each alternative}) = \frac{\epsilon_j}{4}. \quad (19)$$

Thus ϵ_j is the effective per-interface probability of departing from the selected parent event under this template. It is not a universal PbI₂ fault density. The transition matrix propagates these events over multiple layer separations and calculates the phase-weighted pair correlation for each (h, k) rod. The complete six-state construction, its exact rod-dependent reduction, and the finite-stack derivation are given in the Supporting Information.[14, 31–33]

The main physical quantity is the correlation between trilayers separated by Δ interfaces. Let $C_{hk}^{(j)}(\Delta; L, \lambda_s)$ be the disorder-averaged scattering correlation, averaged over all layer pairs at that separation, for population j . Let N be the number of trilayers, d their normal separation, and $q_z = 2\pi L/c$ the crystallite-frame momentum-transfer component along the rod. The corresponding finite-stack rod intensity can be written as

$$S_{hk}^{(j)}(L, \lambda_s) = S_{\text{self},hk}^{(j)}(L, \lambda_s) + \frac{2}{N} \text{Re} \sum_{\Delta=1}^{N-1} (N - \Delta) C_{hk}^{(j)}(\Delta; L, \lambda_s) e^{iq_z d \Delta}. \quad (20)$$

The first term is the intensity scattered by the individual trilayers. The second is their interference, grouped by separation Δ . An ordered parent preserves a periodic correlation over long separations and gives sharp maxima with finite-stack fringes. Fault events dephase or shorten that periodic correlation and transfer intensity into the diffuse regions between the maxima.

The 2H-, 4H-, and 6H-rich populations are treated as mutually incoherent, so their intensities are combined as

$$S_{hk}(L, \lambda_s) = \sum_{j \in \{2H, 4H, 6H\}} w_j S_{hk}^{(j)}(L, \lambda_s), \quad w_{2H} + w_{4H} + w_{6H} = 1. \quad (21)$$

This population-weighted $S_{hk}(L, \lambda_s)$ replaces the ordered finite-stack rod intensity introduced in Sec. 2.2. The subsequent powder-cylinder formation, mosaic broadening, Ewald selection, optical weighting, detector mapping, and finite-region integration are unchanged. The stacking model is therefore a structural module within the same detector-space forward calculation rather than a separate diffraction framework.

5.2 Hierarchy of stacking-disorder models

The selected comparisons should be organized as a nested model hierarchy rather than as a generic low-to-high disorder series. The first stage establishes a nearly ordered 2H baseline. The second asks whether faults within a 2H-rich population alone reproduce the diffuse redistribution. The third adds a 6H-rich population, and the fourth adds a 4H-rich population. Each stage must use the same detector geometry, background treatment, integration regions, and ordered baseline, and added complexity is retained only when it improves the fault-sensitive rod profiles without degrading the ordered maxima.

Hierarchy of PbI_2 stacking-disorder models

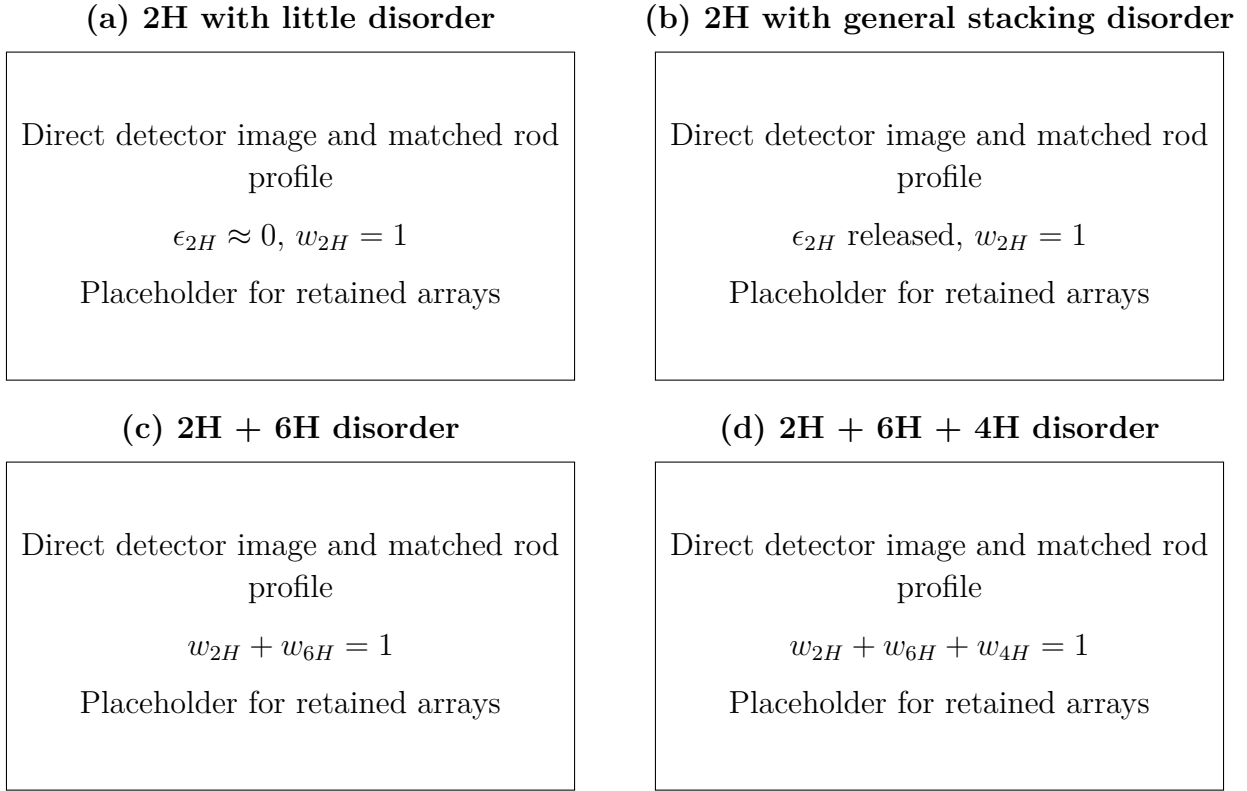


Figure 12: Planned matched comparisons for a nested hierarchy of PbI_2 stacking-disorder models. (a) Nearly ordered 2H establishes the finite-stack baseline. (b) Faulted 2H tests whether one parent-rich population accounts for the diffuse redistribution. (c) A 6H-rich population is added to the 2H model. (d) A 4H-rich population is then added to the 2H+6H model. Every stage will use the same detector projection, background treatment, and intensity normalization. The panels remain placeholders until the direct measured and calculated arrays are inserted.

5.3 Connection to the detector calculation

The stacking parameters are introduced only after detector geometry, sample alignment, mosaicity, lattice constants, and the ordered PbI_2 rod intensity have been fixed by the ordered-film analysis. The core parameters are the departure probabilities ϵ_{2H} , ϵ_{4H} , and ϵ_{6H} , the normalized population weights w_{2H} , w_{4H} , and w_{6H} , and the finite stack length N . Measured and calculated detector images are projected through identical constant- Q_R masks and reported versus Q_z or $L_{\text{proj}} = Q_z c / (2\pi)$. Independent peak areas are not introduced before the stacking calculation because they would absorb the between-peak intensity that constrains the layer correlations.

The hierarchy in Fig. 12 is intended to identify the minimum structural complexity required by the fault-sensitive rods. The nearly ordered 2H model establishes the baseline. Releasing ϵ_{2H} tests general disorder within that parent. The 6H- and 4H-rich populations are then introduced sequentially rather than simultaneously. Numerical transition parameters and population weights should be reported only after the direct experimental profiles have been refit with this hierarchy. The absence of comparable diffuse contrast on $r = 0$ does not imply that those scattering volumes contain no faults. It follows from the phase-insensitive condition described above.

6 Main takeaways

- A two-dimensional powder retains radial and transverse detector information that separates specular-family, off-specular, wavelength-sensitive, and mosaic-sensitive observables more effectively than a conventional one-dimensional reduction.
- The forward model begins with continuous rod intensities $S_{hk}(L, \lambda_s)$, distributes them through the two-dimensional-powder and crystallite-orientation densities, restricts the resulting reciprocal-space density to each state-specific Ewald surface, and maps it to the calibrated detector with the same finite integration used for the data.
- In the Bi_2Se_3 $r = 0$ region, the long 003 extension is radial in 2θ and is reproduced by the wavelength-resolved calculation, while the transverse ϕ width remains the cleaner crystallite-tilt constraint.
- The bright near-origin $r = 0$ response motivates a separate Parratt optical branch, with Névot–Croce roughness, that is smoothly connected to the finite-stack kinematic branch. In the present work this construction distinguishes the near-critical response from the Bragg and mosaic contributions. It is not used as an independent reflectivity refinement.
- PbI_2 stacking faults act by changing trilayer pair correlations within the continuous rod intensity. Off-specular rods retain basal-plane phase contrast and develop diffuse

between-peak intensity, whereas $r = 0$ is contrast-matched to the included stacking events and can contain faults without displaying comparable diffuse contrast.

- The PbI_2 comparison should test a nested hierarchy of nearly ordered 2H, faulted 2H, 2H+6H, and 2H+6H+4H while keeping detector geometry, mosaicity, ordered baseline, projection, and background treatment fixed.
- The staged refinement order is essential: instrumental geometry is determined first, sample alignment second, wavelength and mosaic effects third, ordered rod intensity next, and stacking-disorder parameters last. This ordering limits compensation among parameters and keeps the structural interpretation tied to matched detector-space observables.
- Detector-space indexing establishes where allowed reflections should appear. The present contribution is the subsequent continuous intensity and profile calculation that combines those locations with source phase space, rod intensity, orientation density, optical weights, and finite detector integration.[27]

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